

Negative Hypergeometric Distribution:

Let x be a random variable that denotes the number of trials required to get a fixed number of objects with specified characteristic is called negative hypergeometric random variable and its distribution is called negative hypergeometric distribution.

Definition:

A random variable x is said to be followed a negative hypergeometric distribution with three parameters N, M and r and its pdf is;

$$p(x=x) = \frac{\binom{x-1}{r-1} \binom{N-x}{M-r}}{\binom{N}{M}}, \quad x=r, r+1, r+2, \dots, N$$

Derivation of NHGD:

Let a population consists of N objects out of which M objects have a certain characteristic is called success (say defective) and $(N-M)$ objects do not have the characteristic called failure (say non-defective).

Let x be a random variable that denotes the number of draws needed to draw r th success in sampling without replacement. If the draws are made until r th success is obtained then the r th success is obtained in the last x th draw with probability $\frac{M-(r-1)}{N-(x-1)}$ and $(r-1)$ successes are obtained in the previous draws with probability $\frac{\binom{M}{r-1} \binom{N-M}{x-r}}{\binom{N}{x-1}}$ where $x \geq r$

Therefore, the probability that x draws are needed to get the r th success is given by;

$$P(X=x) = \frac{\binom{M}{r-1} \binom{N-M}{x-r}}{\binom{N}{x-1}} \times \frac{M-(r-1)}{N-(x-1)}$$

$$P(X=x) = \frac{\binom{x-1}{r-1} \binom{N-x}{M-r}}{\binom{N}{M}}, \quad x=r, r+1, r+2, \dots, N.$$

Moments:

Let $x \sim NH(N, M, r)$ then its pmf is

$$P(X=x) = \frac{\binom{x-1}{r-1} \binom{N-x}{M-r}}{\binom{N}{M}}$$

$$\text{Mean, } E(X) = \sum_{x=r}^N x P(X=x)$$

$$= \sum_{x=r}^N x \frac{\binom{x-1}{r-1} \binom{N-x}{M-r}}{\binom{N}{M}}$$

$$= \binom{N}{M} \sum_{x=r}^N x \cdot \frac{\binom{x-1}{r-1} \binom{N-x}{M-r}}{\binom{N}{M}}$$

$$= \binom{N}{M} \sum_{x=r}^N \frac{x(x-1)!}{(r-1)!(x-1-r+1)!} \times \binom{N-x}{M-r}$$

$$= \frac{r}{\binom{N}{M}} \sum_{x=r}^N \frac{x(x-1)!}{(x-r)!(r-1)!} \times \binom{N-x}{M-r}$$

$$= \frac{r}{\binom{N}{M}} \sum_{x=r+1}^N \frac{x!}{(x-r)!r!} \times \binom{N-x}{M-r}$$

$$= \frac{r}{\binom{N}{M}} \left[\sum_{x=r+1}^N \binom{x}{r} \binom{N-x}{M-r} \right]$$

$$= \frac{r}{\binom{N}{M}} \times \binom{N+1}{M+1}$$

$$\begin{aligned}
 &= \frac{r}{N!} \times \frac{(N+1)!}{(M+1)!(N+1-M-1)!} \\
 &= \frac{r \times M! (N-M)!}{N!} \times \frac{(N+1)!}{(M+1)!(N-M)!} \\
 &= r \times \frac{M!}{M!} \times \frac{(N+1)!}{(M+1)M!}
 \end{aligned}$$

$$E(x) = \frac{r(N+1)}{(M+1)}$$

For variance, $V(x) = E(x^2) - [E(x)]^2$

$$\begin{aligned}
 \text{Here, } E(x^2) &= E[x(x+1) - x] \\
 &= E[x(x+1)] - E(x) \\
 &= \sum_{x=1}^N x(x+1) p(x=x) - \sum_{x=1}^N x p(x=x)
 \end{aligned}$$

$$= \sum_{x=1}^N x(x+1) \frac{(x-1)! \binom{N-x}{r-1}}{\binom{N}{M}} - \frac{r(N+1)}{(M+1)}$$

$$= \frac{1}{\binom{N}{M}} \sum_{x=1}^N x(x+1) \frac{(x-1)!}{(r-1)!(x-1-r+1)!} \binom{N-x}{M-r} - \frac{r(N+1)}{(M+1)}$$

$$= \frac{1}{\binom{N}{M}} \sum_{x=1}^N x(x+1) \frac{(x-1)!}{(r-1)!(x-1)!} \binom{N-x}{M-r} - \frac{r(N+1)}{(M+1)}$$

$$= \frac{r(r+1)}{\binom{N}{M}} \sum_{x=r+2}^N \frac{(x+1)x(x-1)!}{(x+1)r(r-1)!(x-r)!} \binom{N-x}{M-r} - \frac{r(N+1)}{(M+1)}$$

$$= \frac{r(r+1)}{\binom{N}{M}} \sum_{x=r+2}^N \frac{(x+1)x(x-1)!}{(x+1)r(r-1)!(x-r)!} \binom{N-x}{M-r}$$

$$= r \frac{N+1}{M+1}$$

$$= \frac{r(r+1)}{\binom{N}{M}} \left[\sum_{x=r+2}^N \binom{x+1}{r+1} \binom{N-x}{M-r} \right] - r \frac{N+1}{M+1}$$

$$E(x^2) = \frac{r(r+1)}{\binom{N}{M}} \left\{ \frac{N+2}{M+2} \right\} - r \frac{N+1}{M+1}$$

$$E(x^2) = \frac{r(r+1)}{N!} \times \frac{(N+2)!}{(M+2)!(N+2-M-2)!} - r \frac{N+1}{M+1}$$

$$= \frac{r(r+1) M! (N-M)! (N+2)!}{N! (M+2)!(N-M-2)!} - r \frac{(N+1)}{(M+1)}$$

$$= \frac{r(r+1) N!}{N!} \times \frac{(N+2)(N+1) N!}{(M+2)(M+2) N!} - r \frac{(N+1)}{(M+1)}$$

$$E(x^2) = \frac{r(r+1) (N+2)(N+1)}{(M+2)(M+2)} - r \frac{(N+1)}{(M+1)}$$

$$\therefore V(x) = E(x^2) - [E(x)]^2$$

$$= \frac{r(r+1)(N+2)(N+1)}{(M+2)(M+2)} - \frac{r(N+1)}{(M+1)} - \frac{r^2(N+1)^2}{(M+1)^2}$$

$$= \frac{r(N+1)}{(M+1)} \left[\frac{(r+1)(N+2)}{M+2} - 1 - \frac{r(N+1)}{M+1} \right]$$

$$V(x) = \frac{r(N+1)(N-M)(M+1-r)}{(M+1)^2(M+2)}$$

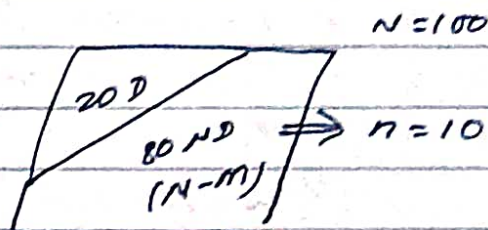
Example:

A lot consists of 100 bulbs of which 20 bulbs are defective. The lot is inspected by taking random sample of 10 bulbs & testing each of the bulbs in the sample. If the number of defective bulbs is at most 1, the lot is accepted; otherwise the lot is rejected. Find the prob. of accepting the lot.

Soln:

$$P(X \leq 1) = ?$$

$$P(X \leq 1) = P(X=0) + P(X=1)$$



Here, \$X \sim \text{HGD}\$, then its prob. is;

$$P(X=x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}, \quad x = 0, 1, 2, \dots, \min(n, M)$$

$$P(X=0) = \frac{\binom{20}{0} \binom{100-20}{10-0}}{\binom{100}{10}} =$$

$$P(X=1) = \frac{\binom{20}{1} \binom{100-20}{10-1}}{\binom{100}{10}} =$$

Example:

A lot contain 10 items of which 3 items are defective. If items are drawn at random, one at a time without replacement from the lot

(i) Find the prob. of getting second defective items in the 6th draw.

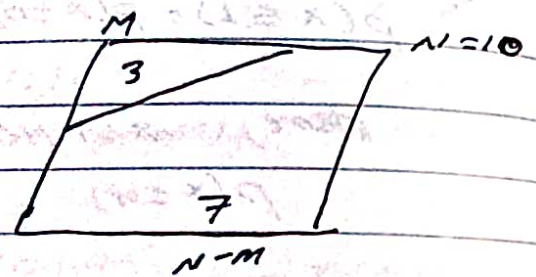
(ii) Find the prob. that at most 3 draws are needed to get 2 defective items.

Soln:

(i) $N=10$

$M=3$

$r=2$



$P(X=6) = ?$

Here,

$X \sim NHG$, then its pmf is;

$$P(X=x) = \frac{\binom{x-1}{r-1} \binom{N-x}{M-x}}{\binom{N}{M}}$$

$$P(X=6) = \frac{\binom{6-1}{2-1} \binom{10-6}{3-2}}{\binom{10}{3}} = \frac{\binom{5}{1} \binom{4}{1}}{\binom{10}{3}}$$

$$= \frac{5 \times 4}{120} = \frac{1}{6} = 0.167$$

(ii) $P(X \leq 3) = \sum_{x=1}^3 P(X=x)$

For, $r=2$

$$P(X \leq 3) = \sum_{x=2}^3 P(X=x)$$

$$= P(X=2) + P(X=3)$$

Here, $X \sim NHG$, then its prob is;

$$P(X=x) = \frac{\binom{x-1}{r-1} \binom{N-x}{M-r}}{\binom{N}{M}}$$

$$P(X=2) = \frac{\binom{2-1}{2-1} \binom{10-2}{3-2}}{\binom{10}{3}} = \frac{\binom{1}{1} \binom{8}{1}}{\binom{10}{3}}$$

$$= \frac{1 \times 8}{120} = \frac{1}{15} = 0.067$$

$$P(X=3) = \frac{\binom{3-1}{2-1} \binom{10-3}{3-2}}{\binom{10}{3}} = \frac{\binom{2}{1} \binom{7}{1}}{\binom{10}{3}}$$

$$= \frac{2 \times 7}{120} = \frac{7}{60} = 0.1167$$

$$\therefore P(X \leq 3) = P(X=2) + P(X=3)$$

$$\text{or, } P(X \leq 3) = 0.067 + 0.1167$$

$$\Rightarrow P(X \leq 3) = 0.1837$$

Example: An urn contains 7 white and 3 black balls. 5 balls are drawn from the urn. What is the probability that balls drawn are white.

① If the balls are drawn with replacement

② If the balls are drawn without replacement

Solⁿ:

① Here:

$$P(W) = 7/10 \Rightarrow p = 7/10$$

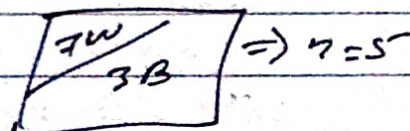
$$P(B) = 3/10 \Rightarrow q = 3/10$$

$$n = 5$$

$$P(X=5) = ?$$

Here,

$X \sim B(n, p)$ then its prob is;



$$P(X=x) = {}^n C_x p^x q^{n-x}, \quad x=0,1,2,\dots,n$$

$$P(X=x) = {}^5 C_5 \left(\frac{7}{10}\right)^5 \left(\frac{3}{10}\right)^{5-5}$$

$$= 1 \times 0.16807 \times 1$$

$$= 0.16807$$

$$(11) \quad P(X=5) = ?$$

$$\text{Here, } X \sim HG(N, M)$$

$$P(X=x) = \frac{{}^M C_x {}^{N-M} C_{n-x}}{{}^N C_n}$$

$$P(X=5) = \frac{{}^7 C_5 {}^{10-7} C_{5-5}}{{}^{10} C_5}$$

$$= \frac{21 \times 1}{252}$$

$$= 1/12$$

$$= 0.0833$$

Example: From a class containing 20 smokers and 80 non-smokers, 10 students are chosen at random without replacement.

(1) Obtain the prob. of finding exactly 5 smokers and 5 non-smoker.

(2) Obtain mean and variance of the number of smokers.

Solⁿ:

Here,

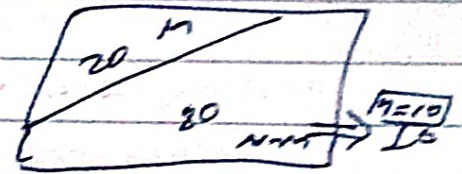
$$P(X=x) = ?$$

Here, $X \sim HG(N, M)$ then

its pmf is;

$$P(X=x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$$

$$P(X=5) = \frac{\binom{20}{5} \binom{80}{10-5}}{\binom{100}{10}}$$



$$x=5$$

$$N-M=80$$

$$N-20=80$$

$$N=100$$

$$= \frac{15504 \times 24040016}{1.731030946 \times 10^{13}}$$

$$= 0.02153$$

$$(ii) E(X) = \frac{NM}{N} = \frac{10 \times 20}{100} = 2$$

$$V(X) = \frac{NM(N-M)(N-n)}{N^2(N-1)}$$

$$= \frac{10 \times 20 (100-20) (100-10)}{100^2 (100-1)}$$

$$= \frac{200 \times 80 \times 90}{100^2 \times 99}$$

$$= \frac{16}{11}$$

$$= 1.4545$$

Example: A lot contains 15 items of which 4 are defectives. If items are drawn at random one at a time without replacement from the lot. Find

(i) the prob. of getting second defective item on the 6th draw.

(ii) the mean and variance of the distribution.

Soln:

$$N=15, M=9, r=2, x=6$$

Here, $X \sim \text{NHD}(N, M, r)$, then its pmf is

$$P(X=x) = \frac{\binom{x-1}{r-1} \binom{N-x}{M-r}}{\binom{N}{M}}, \quad x=r, r+1, r+2, \dots, N$$

$$= \frac{\binom{6-1}{2-1} \binom{15-6}{9-2}}{\binom{15}{9}} = \frac{\binom{5}{1} \binom{9}{2}}{\binom{15}{4}} = \frac{12}{91} = 0.1318$$

$$(i) E(X) = r \frac{N+1}{M+1} = 2 \frac{15+1}{4+1} = 2 \times \frac{16}{5}$$

$$\therefore E(X) = 6.4$$

$$V(X) = \frac{r(N+1)(N-M)(M+1-r)}{(M+1)^2(M+2)}$$

$$= \frac{2(15+1)(15-9)(9+1-2)}{(4+1)^2(4+2)}$$

$$= \frac{2 \times 16 \times 6 \times 3}{5^2 \times 6}$$

$$= \frac{176}{25}$$

$$= 7.04$$